This slowing of time near a black hole plays a major role in *Interstellar*. Cooper despairs of ever seeing his daughter Murph again, when his travel near Gargantua causes him to age only a few hours while Murph, on Earth, is aging eight decades.

Human technology was too puny to test Einstein's law until nearly half a century after he formulated it. The first good test came in 1959 when Bob Pound and Glen Rebca used a new technique called the Mössbauer effect to compare the rate of flow of time in the basement of a 73-foot tower at Harvard University with time in the tower's penthouse. Their experiment was exquisitely accurate: good enough to detect differences of 0.0000000000016 seconds (1.6 trillionths of a second) in one day. Remarkably, they found a difference 130 times larger than this accuracy and in excellent agreement with Einstein's law: Time flows more slowly in the basement than in the penthouse by 210-trillionths of a second each day.

The accuracy improved in 1976, when Robert Vessot of Harvard flew an atomic clock on a NASA rocket to a 10,000-kilometer height, and used radio signals to compare its ticking rate with clocks on the ground (Figure 4.1). Vessot found that time on the ground flows more slowly than at a height of 10,000 kilometers by about 30 microseconds (0.00003 seconds) in one day, and his measurement agreed with Einstein's law of time warps to within his experimental accuracy. That accuracy (the uncertainty in Vessot's measurement) was seven parts in a hundred thousand: 0.00007 of 30 microseconds in a day.